

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)							Semester: II		
Course: Fundamentals of Computer Systems and Networking							Code: ECES204		
Teaching Scheme (Hrs/week)				Evaluation Scheme (Marks)					
Lecture	Practical	Tutorial	Credit	CIE	ETE	TW	OR	PR	Total
02	02	-	03	40	60	25	-	-	125
Prerequisites:									
Basic knowledge of computers and binary systems.									
Course Objectives:									
1. To understand the architecture and functioning of computer systems.									
2. To explore fundamental networking concepts and technologies.									
3. To develop foundational knowledge of operating systems and computer organization.									
4. To learn about various networking models, protocols, and data communication methods.									
5. To understand the role of hardware and software in computing and networking.									
Course Outcomes: After completion of this course, students will able to -									
CO1	Understand the basic components and organization of a computer system and the role of operating systems in managing hardware and software.								
CO2	Gain insights into the organization and architecture of a computer, including CPU functioning and memory hierarchy.								
CO3	Understand basic networking concepts, data communication modes, network topologies, and the types of networks.								
CO4	Describe the OSI and TCP/IP models, along with understanding key networking protocols and addressing techniques.								
CO5	Understand the basic concepts of network security, including encryption, firewalls, and security protocols to protect communication.								
CO6	Explore the emerging trends in computer systems and networking, including cloud computing, IoT, and advancements in network technologies								
Course Contents:									
Unit	Description								Duration (Hrs.)
1.	Introduction to Computer Systems: Overview of Computers: History of computers (evolution and key milestones), Types of computers: Analog, Digital, And Hybrid. Applications of Computers: In education, healthcare, business, entertainment, and other fields. Components of a Computer System: Hardware vs. Software, Basic hardware components (CPU, memory, storage, input/output devices). Basic Organization of a Computer: CPU (Control Unit, Arithmetic Logic Unit, Registers), Memory hierarchy (Primary, Secondary, Cache, Virtual). Data Representation: Number systems (Binary, Octal, Decimal, Hexadecimal), Binary arithmetic (addition, subtraction, multiplication, division), Character representation (ASCII, Unicode).								4

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	Introduction to Operating Systems: Functions (process, memory, file system, device management), Types (batch, time-sharing, real-time, distributed, embedded), Structure (Kernel, Shell, System Utilities).	
2.	Computer Architecture and Organization: Basic Structure of a Computer: Von Neumann architecture, instruction cycle. CPU Organization: ALU, Registers, Control Unit. Buses and Interfacing: Overview of data transfer methods (bus organization, control lines). Memory Architecture: RAM, ROM, Cache, Virtual Memory. I/O Systems: I/O devices, I/O addressing, and basic concepts of interrupts and DMA. Introduction to Graphics Processing Unit (GPU): Understand the functionalities of a GPU in graphics processing and video acceleration. Display Technologies: Explore different display types (CRT, LCD, LED), display technology fundamentals, resolution, and refresh rate	4
3.	Data Communication and Networking Fundamentals: Data Communication: Types of data transmission, Modes (Simplex, Half-duplex, Full-duplex), Transmission media (Wired, Wireless) Network Topologies: Bus, Star, Ring, Mesh, Hybrid. Types of Networks: LAN, WAN, MAN, PAN Overview of Networking Devices: Switches, Routers, Modems, Hubs, Repeaters Introduction to the Internet: Basics of how the internet works, Internet protocols	5
4.	Networking Models and Protocols: OSI Model: Layers and functions. TCP/IP Model: Layers and comparison with OSI. IP Addressing: IPv4, IPv6, Subnetting, CIDR. Network Protocols: HTTP, FTP, SMTP, DNS, DHCP, ICMP. Packet Switching vs. Circuit Switching: Fundamental differences and use cases.	5
5.	Introduction to Network Security: Fundamentals of Network Security: Threats, Attacks, Vulnerabilities. Cryptography Basics: Symmetric and Asymmetric Encryption, hashing. Firewalls and Intrusion Detection Systems (IDS): Working principles of firewalls and intrusion detection systems. Security Protocols: SSL/TLS, IPsec, VPN. Authentication and Access Control: Overview of methods such as passwords, biometrics, multi-factor authentication.	5
6.	Emerging Trends in Computer Systems and Networking: Cloud Computing: Basics, service models (IaaS, PaaS, SaaS), deployment models (public, private, hybrid). Virtualization: Concept of virtual machines, hypervisors. Internet of Things (IoT): Architecture, Applications, Challenges. 5G Networking: Features, use cases, and its role in modern communications. Edge and Fog Computing: Concepts and Applications. Data Centers: Fundamentals, architectures, and their role in modern computing	5
TOTAL		28

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List of Experiments:

Group A: Fundamentals of Computer Systems: (Any 8)

1. Disassemble and identify key components of a computer system (CPU, RAM, motherboard, storage, etc.). Discuss functionalities and basic maintenance practices.
2. Use a simulator to demonstrate the instruction execution process in a CPU and explore the memory hierarchy (RAM, cache, virtual memory). **Open-source software:** SimulIDE, Little Man Computer Simulator.
3. Create a diagram of a motherboard, labeling key components (CPU socket, RAM slots, expansion slots, connectors) and explaining their functions. **Open-source software:** [Dia](#), [Fritzing](#).
4. Install and configure an expansion card (e.g., graphics card or network card) in a computer.
5. Compare HDD and SSD by conducting performance tests (e.g., read/write speed tests). **Open-source software:** CrystalDiskMark, [KDiskMark \(Linux\)](#)
6. Install an operating system (e.g., Windows, Linux) on a computer. **Open-source software:** [Ubuntu](#), [Fedora](#).
7. Benchmark a GPU using a graphics-intensive application and analyze its performance. **Open-source software:** Unigine Heaven Benchmark, GLMark2
8. Compare different display technologies (CRT, LCD, LED) in terms of resolution, refresh rate, and overall quality.
9. Diagnose and resolve a hardware or software problem in a computer system. **Open-source software:** [HWiNFO](#), Speccy
10. Install and configure antivirus software, demonstrating its features. **Open-source software:** [ClamAV](#).

Group B: Fundamentals of Networking: (Any 5)

1. Set up a small LAN and demonstrate data transfer between devices. **Open-source software:** [Wireshark](#), EtherApe
2. Configure a switch and a router for a network, demonstrating their roles in data communication. **Open-source software:** Cisco Packet Tracer, [GNS3](#)
3. Simulate data transfer using the TCP/IP model and analyze packet data. **Open-source software:** [Wireshark](#)
4. Create a presentation on common malware and viruses, including preventive measures and real-world examples.
5. Research and create a report on different data center topologies and architectures (e.g., star, mesh, tree).
6. Conduct a case study of a specific data center or take a virtual tour of a data center, highlighting key components and their functions.

Text Books:

1. David A. Patterson and John L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface," Morgan Kaufmann, 2017.
2. Behrouz A. Forouzan, "Data Communications and Networking," McGraw-Hill, 2017.
3. James F. Kurose and Keith W. Ross, "Computer Networking: A Top-Down Approach," Pearson, 2020.
4. William Stallings, "Cryptography and Network Security: Principles and Practice," Pearson, 2017.

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5. Thomas Erl, "Cloud Computing: Concepts, Technology & Architecture," Prentice Hall, 2013.

Reference Books:

1. Andrew S. Tanenbaum, "Modern Operating Systems," Pearson, 2015.
2. M. Morris Mano, "Computer System Architecture," Pearson, 2013.
3. Douglas E. Comer, "Internetworking with TCP/IP," Pearson, 2018.
4. Jerome H. Saltzer and M. Frans Kaashoek, "Principles of Computer System Design: An Introduction," Morgan Kaufmann, 2009.

E-Resources:

1. <https://nptel.ac.in/courses/106103068>
2. <https://nptel.ac.in/courses/106105081>
3. <https://nptel.ac.in/courses/106104449>